Memory-Defined Phase Maps for Dissipative Quantum Reservoirs

Anonymous QCE Submission

***Abstract*—Quantum reservoir computing performs temporal machine learning by training only a linear readout on the measured outputs of a fixed quantum system, leaving the quantum dynamics itself untrained. This makes the approach attractive for near-term hardware, but it sharpens a question that has remained largely unresolved: which device settings render the fixed dynamics useful rather than merely high-dimensional. We address this question for a stateful dissipative gate-model reservoir and demonstrate that strong performance does not occupy arbitrary coordinates of the control space. Instead, it concentrates in a well-defined operating region governed by the interplay of input drive, recurrent coupling, and controlled dissipation. This region is transferable: an operating band identified on a subset of tasks predicts effective settings on held-out tasks and held-out reservoir initializations alike. Mechanism ablations establish that the region is causal rather than incidental, and an inexpensive memory-based diagnostic recovers it prior to any task-specific evaluation. The result reframes reservoir selection as the identification of a transferable, memory-preserving regime rather than the pursuit of a single benchmark optimum.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Question and Claim

Reservoir computing trains a linear readout on fixed nonlinear dynamics [1], [2]. In a quantum reservoir, Hilbert-space dimension alone is not the useful resource. The dynamics must preserve recent input history, forget old history, and expose nonlinear observables to the readout [4], [5], [6]. Too much input drive overwrites state, zero damping prevents controlled fading memory, and missing recurrent mixing leaves weak temporal features.

We therefore treat the phase diagram as the scientific object. The claim is a QRC-only generalization claim: for the studied stateful dissipative QRC family, useful learning is organized by a transferable operating regime. The strongest reviewer-safe statement is a hypothesis supported by four tasks, ten seeds, and ablations: scalar temporal tasks that can be learned by this architecture should be searched first in a low-input, nonzero-damping, nonmaximal-coupling band.

II. Protocol

The reservoir carries a density matrix from one input time step to the next. For scalar input u_t,

$$\rho_{t,0} = U_{\mathrm{in}}(u_t)\rho_{t-1,L}U_{\mathrm{in}}^{\dagger}(u_t), \qquad U_{\mathrm{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

Each virtual layer applies frozen local rotations, a ZZ ring, and a product noise channel:

$$\begin{aligned} \rho_{t,\ell} &= \mathcal{D}_{\gamma}^{\otimes n}\left[U_{\lambda}U_{\mathrm{mix}}\rho_{t,\ell-1}U_{\mathrm{mix}}^{\dagger}U_{\lambda}^{\dagger}\right], \\ U_{\lambda} &= \prod_{(i,j)\in E_{\mathrm{ring}}} e^{-i\lambda Z_i Z_j}. \end{aligned} \tag{2}$$

The main phase maps use $n = 4$, two layers, ring topology, uniform input, amplitude damping, the $Z+ZZ$ QRC16 readout, and ridge-only training. Hyperparameters are never selected by holdout data.

For task j and seed s, let $r_{j,s}(\theta)$ be the validation percentile rank of $\theta = (\beta, \lambda, \gamma)$, with lower rank better. A transferable operating band is

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

This frequency-level definition is deliberately weaker than a universal optimum. It asks whether a region repeatedly behaves well, not whether the same coordinate wins every task.

III. Evidence

Phase-map transfer. Fig. 1 shows the core pattern: low validation rank is concentrated, not randomly scattered. The primary band $B_{20,0.7}$ has 4 connected points, all at $\gamma = 0.12$. When one task is withheld while defining the band from the remaining tasks, the held-out task-seed mean validation rank is 0.240 with bootstrap CI $[0.215, 0.265]$. When one seed is withheld, the corresponding mean rank is 0.180 with CI $[0.154, 0.209]$. The band is therefore a top-quartile search prior that transfers across both task identity and reservoir initialization.

Final holdout audit. After $B_{20,0.7}$ is fixed by validation ranks, the selected band is evaluated on the held-out split. Table II reports the mean over the validation-selected band and the fixed medoid; no coordinate is chosen by holdout NMSE. The aggregate band holdout rank is 0.156, so the operating-regime prior remains top-quartile on unseen data.

Mechanism stress tests. The operating band is not just a plotting artifact. Figure 2(a) shows the validation-defined frequency map for the primary band, and Fig. 2(b–d) shows where rank is lost when mechanisms are removed. At the central slice, the projected base band has 4 points. Removing the controlled damping mechanism gives retention 0.00; removing recurrent mixing gives retention 0.00; replacing amplitude damping by dephasing gives retention 0.00. In every focused ablation, the base band ceases to be a robust top-ranked region under the same validation-rank criterion. The result supports

TABLE I

QRC-ONLY AUDIT TRAIL. HOLDOUT DATA ARE NOT USED TO DEFINE THE OPERATING REGIME.

Item	Value
Tasks and splits	Mackey–Glass, Lorenz, NARMA10: washout/train/validation/holdout $= 300/1600/400/400$; annual Sunspots: $20/150/50/60$. Inputs are scaled to $[-1, 1]$ and targets standardized.
Seeds and ridge	Reservoir seeds 42–51; ridge grid $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$ selected on validation.
Phase grid	QRC16 coarse grid over $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$.
Primary band	$B_{20,0.7}$ contains 4 connected points; the medoid is $(\beta/\pi, \lambda/\pi, \gamma) = (0.04, 0.12, 0.12)$. Relaxing to $B_{20,0.6}$ gives 14 points and $B_{30,0.7}$ gives 46 points.
Stress tests	Focused QRC-only ablation maps at $\gamma = 0.12$ compare base amplitude damping against $\gamma = 0$, dephasing, depolarizing, no mixer, Rx-only mixer, and ZZ-only mixer. The readout, tasks, seeds, and ridge protocol are unchanged.
Diagnostics	Memory capacity and intrinsic processing capacities are computed from simulator trajectories and compared to validation rank and top-decile screening retention.

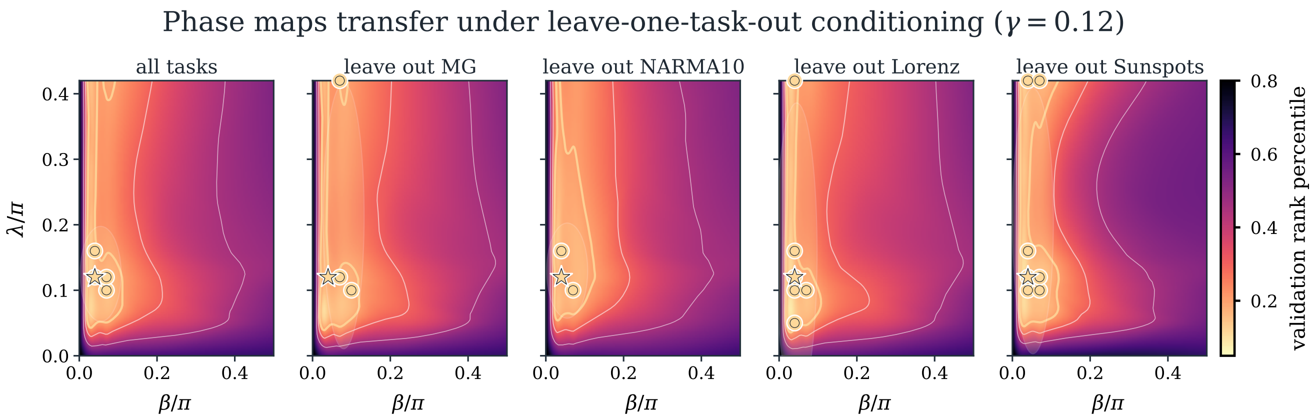

Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. Gold circles mark points that appear in the top-20% for at least 70% of the displayed task-seed replicates; the star is the medoid of the primary band. The same low-input, nonmaximal-coupling region persists under leave-one-task-out conditioning.

TABLE II

HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. MEANS ARE OVER TEN SEEDS PER TASK; "ALL" AVERAGES THE 40 TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	8.77e-05	8.19e-05	0.105
Lorenz	5.28e-05	7.36e-05	0.169
NARMA10	0.561	0.520	0.172
Sunspots	0.194	0.179	0.177
All	0.189	0.175	0.156

the mechanism story: useful dynamics need input that is not too strong, recurrent mixing, and damping that preserves usable memory rather than merely adding noise.

Variant	Band pts.	Retention	Rank on base band
base AD+RxZZ	4	1.00	0.256
$\gamma = 0$	0	0.00	0.465
dephasing	0	0.00	0.442
depolarizing	0	0.00	0.526
no mixer	0	0.00	0.403
Rx only	0	0.00	0.521
ZZ only	0	0.00	0.503

Diagnostics. Memory-based diagnostics identify the same useful region before task-specific holdout evaluation. On the current simulator-backed diagnostic rerun, MC has Spearman $\rho_s = -0.85$ against validation rank; $\mathrm{IPC}_{\mathrm{mem}}$, $\mathrm{IPC}_{\mathrm{tot}}$, and $\mathrm{IPC}_{\mathrm{nonlin}}$ give -0.84, -0.90, and -0.80. Feature variation and effective rank are weakly positive, 0.16 and 0.15, so raw diversity alone is not the mechanism.

IV. DISCUSSION AND SCOPE

The evidence supports a universal operating-regime hypothesis, not a theorem. The tested tasks differ in chaotic smooth dynamics, nonlinear memory, and real annual observations, and the same frequency-defined phase region remains useful across seeds and leave-one-task-out splits. The statement should still be read within the modeled family: four-qubit stateful gate-model QRCs, scalar time-series prediction, exact simulated expectation values, ridge readouts, and the specified grid.

The scope is intentionally narrower than a hardware or advantage claim. Finite-shot sampling, measurement-basis scheduling, calibration drift, and device noise are not modeled. Likewise, the phase map does not say that every task has the same optimum. It says that useful QRC learning should first be searched in a memory-preserving band, and that leaving this band by removing damping or recurrent mixing destroys the robust phase-map signature.

V. CONCLUSION

Dissipative QRC performance is organized by a memory-defined operating regime. The phase maps transfer across tasks

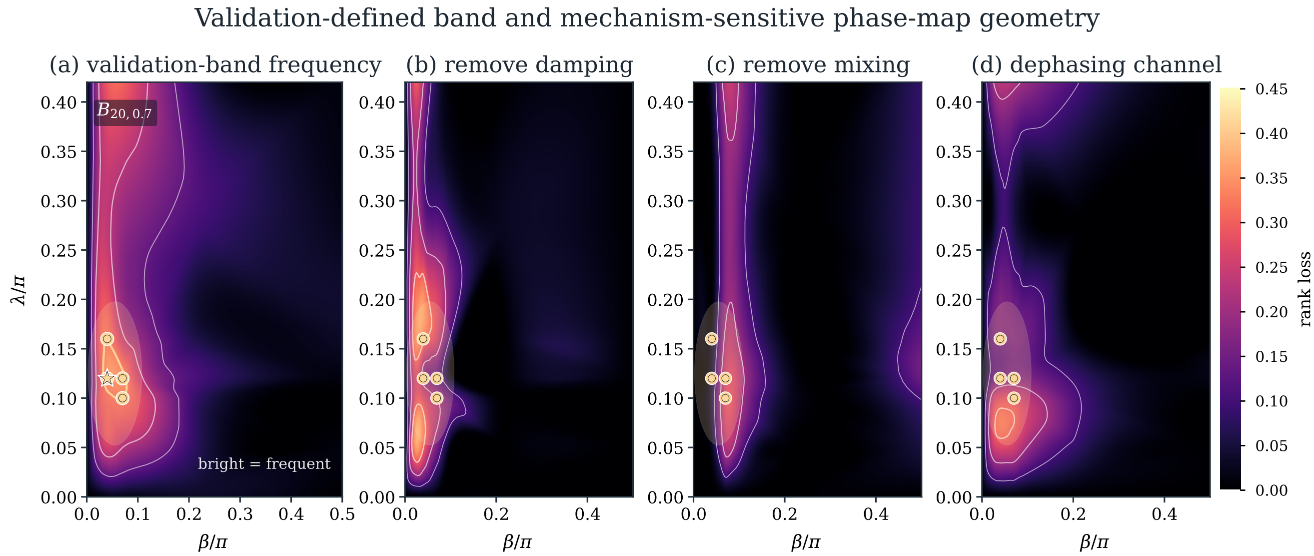

Fig. 2. QRC-only phase-map evidence for the operating-regime hypothesis. (a) The top-20 frequency map visualizes the validation-defined $B_{20,0.7}$ band at $\gamma = 0.12$: bright coordinates repeatedly rank in the top validation quintile across task-seed replicates. (b–d) Ablation rank-loss maps show how much the mean validation rank degrades when controlled damping is removed, recurrent mixing is removed, or amplitude damping is replaced by dephasing. The highlighted base-band projection lies where these mechanism removals cause rank loss.

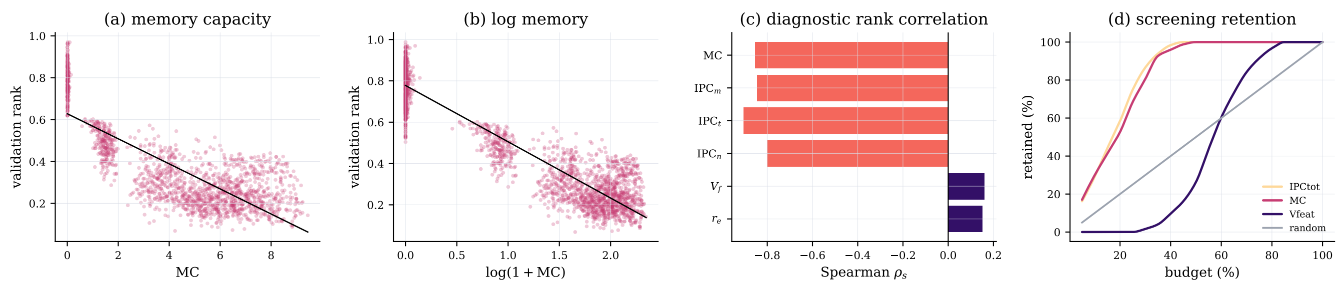

Fig. 3. Diagnostic audit for the QRC phase-map claim. Raw and log-scaled memory capacity both anticorrelate with validation rank, intrinsic processing-capacity diagnostics give the strongest rank correlations, and screening by memory/IPC recovers useful reservoirs earlier than random or diversity-only screens.

and seeds; mechanism ablations remove the robust base band; and intrinsic memory/processing diagnostics recover the same useful region. For this QRC family, the practical claim is simple: before scaling features or chasing a benchmark winner, identify the phase-map band where input drive, recurrence, and damping jointly preserve usable temporal memory.

References

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TABLE III
ARTIFACT MAP FOR EXACT REPRODUCTION OF THE QRC-ONLY PAPER.

Artifact	Role
QRC phase grid	data/qrc_seed_ensemble_grid.csv: main QRC16 phase-map grid over tasks, seeds, and (β, λ, γ).
QRC ablation slices	data/qrc_phase_ablation_slice_grid.csv: focused mechanism stress tests at $\gamma = 0.12$.
Band and holdout statistics	data/phase_map_generalization_stats.json and data/phase_map_holdout_performance.csv: band sizes, connectedness, leave-one-task/seed bootstrap CIs, post-selection holdout audit, ablation retention, and diagnostic correlations.
Generated numbers	paper/generated/phase_map_numbers.tex: all numeric macros used in the manuscript.
Figures	scripts/make_figures_and_build_data.py: regenerates the phase maps, band-frequency view, ablation rank-loss maps, and diagnostic screening plots.
Build	./reproduce.sh and ./paper/build.sh --update-pdf: end-to-end regeneration and PDF refresh.